Factorization memory in a conic ideal: the A_3, B_3, and H_3 configurations

Tavis Rudd

Working split draft — July 2026

Abstract

Restriction to a conic forgets how its marked points were paired into secants. We study the information retained after differences of matching products are divided by the conic equation. We prove that this quotient is intrinsic up to translation and equivariant under the endpoint stabilizer. For the rank-three Coxeter configurations A_3, B_3, and H_3, its difference space has ranks 3, 6, and 10. Uniformly, the image contains the top harmonic summand and, in even degree, the deepest radial line. In the H_3 case it omits exactly the middle Fischer layer $Q\mathcal{H}_2$. In the balanced cases, second moments recover two complementary sheets, a signed cubic orients them, and six incidence profiles recover the matching-table row. A modular depth quotient and arithmetic gluing describe how this factorization memory changes across characteristics.

Keywords. conic ideal; secant matching; Coxeter configuration; factorization; cubic invariant; modular representation.

2020 Mathematics Subject Classification. 05B25, 51E22, 20C20, 13A02.

1 Introduction

The companion rigidity paper isolates the Clebsch hexagon through its uncovered syndrome locus and reconstructs its projective geometry from decoding data. Here we begin with marked secant geometry and ask a complementary reconstruction question: what does common restriction to a conic forget, and what does the conic-ideal quotient remember? We answer this question across the rank-three A_3, B_3, and H_3 configurations.

Theorem 1.1 (Factorization recovery). *Let the A_3, B_3, and H_3 matching configurations be the finite matching orbits defined in Section 4. Then:*

(i) *canonically scaled matching products have equal restriction to the conic, and their conic-ideal quotients have difference ranks* 3*,* 6*, and* 10*;*

(ii) *in the B_3 and H_3 cases, the two factorization sheets are, up to interchange, the unique complementary halves with equal first and second tensor moments;*

(iii) *the signed moments vanish through degree two, while a nonzero degree-three tensor transforms by the character that exchanges the sheets;*

(iv) *in the H_3 case, six explicit incidence profiles separate the six A_4–A_5 double-coset labels, and the singleton profiles recover the corresponding decorated matching-table row.*

The theorem follows one mechanism from a four-endpoint switch to the recovery of finite matching data. Division by the conic equation first turns different factorizations of one restricted form into an affine configuration. Quadratic products then recover an unordered bipartition; the first signed moment that survives is cubic. Finally, a double-coset incidence map compresses the configuration to six profiles without identifying any two labels. The rigidity theorem motivates this question but is not a hypothesis: all marked-conic objects and quotient constructions are defined independently here.

This line of research arose from the following normal-play placement game. Two players alternately adjoin a point of a finite projective plane while keeping the selected set an arc; the first player with no legal point loses. After frame reduction in an odd projective plane, a residual three-point position and the two burned directions determine a five-arc and hence a conic; adjoining an on-conic response exposes the exceptional six-arc geometry studied here. The resulting connection now runs in both directions. The small-arc equality theorem isolates the projective frame over $\mathbb{F}_5$ and the Clebsch hexagon over $\mathbb{F}_{11}$ as the only configurations of sizes four through seven whose uncovered locus is a full conic. The latter yields an icosahedral continuation complex, a certified game position, and a symmetry-compressed finite model for constructing second-player responses. Under the standard arc–MDS–AME dictionaries, the same H_3 six-arc gives an $[6, 3, 4]_{11}$ MDS code and a minimum-support stabilizer absolutely maximally entangled state on six 11-level systems, so its projective rigidity controls a concrete quantum-code input. More generally, prescribed-hole arc theory places its equality and reconstruction properties in a forward/inverse framework, while the syndrome–uncovered-point mechanism extends to projective Reed–Solomon deep holes beyond redundancy four. These are applications and continuations of the geometry, not hypotheses of the results proved here.

2 Conic matching products

Write the standard conic as

$$\mathcal{C} = V(Q), \qquad Q = XZ - Y^2,$$

and parametrize it by $\nu(s : t) = (s^2 : st : t^2)$. For points $a = (a_0 : a_1)$ and $b = (b_0 : b_1)$ of $\mathbb{P}^1$, put $[a, b] = a_0b_1 - a_1b_0$. We scale the secant L_{ab} through $\nu(a)$ and $\nu(b)$ by the condition

$$L_{ab}(\nu(s : t)) = [a, (s : t)]\,[b, (s : t)].$$

Let Ω be a set of $2m$ distinct marked points of $\mathbb{P}^1$, with fixed vector representatives. For a perfect matching M of Ω, define its matching product

$$P_M = \prod_{\{a,b\}\in M} L_{ab}.$$

The vector representatives are part of the marked-conic data. Their role is to make the products comparable before passing to projective equivalence.

Proposition 2.1 (The matching-secant quotient)**.** *For every $m \geq 2$ and every two perfect matchings M, N of Ω,*

$$P_M|_{\mathcal{C}} = P_N|_{\mathcal{C}}, \qquad P_M - P_N \in (Q).$$

Consequently, after choosing a reference matching M_0,

$$\Phi_{M_0}(M) = \frac{P_M - P_{M_0}}{Q} \in H^0(\mathbb{P}^2, \mathcal{O}(m-2)) \tag{2.1}$$

is an affine configuration. Changing the reference matching translates this configuration. If $g \in \mathrm{GL}_2$ preserves Ω, $\rho(g)$ is its action on binary quadratics, and representatives are transported by g, then

$$\Phi_{gM_0}(gM) \circ \rho(g) = \det(g)^{2m-2} \Phi_{M_0}(M). \tag{2.2}$$

Thus the difference space, its rank, and its affine moment conditions do not depend on M_0 and are equivariant under the endpoint stabilizer.

Proof. For $x \in \mathbb{P}^1$,

$$P_M(\nu(x)) = \prod_{a \in \Omega} [a, x],$$

which does not depend on the pairing. The kernel of the Veronese pullback

$$k[X, Y, Z] \longrightarrow k[s, t], \qquad (X, Y, Z) \longmapsto (s^2, st, t^2)$$

is the principal ideal (Q), so Q divides every difference. Changing the reference follows by subtraction. Finally, $L_{ga,gb} \circ \rho(g) = \det(g)^2 L_{ab}$ and $Q \circ \rho(g) = \det(g)^2 Q$, which gives (2.2). □

3 Switches and divisibility

For four distinct endpoints, the Plücker relation gives the local switch identity

$$L_{ab}L_{cd} - L_{ac}L_{bd} = [a, d][b, c]\, Q. \tag{3.1}$$

Multiplying (3.1) by the secant factors common to two matchings proves divisibility for a single switch. Any two perfect matchings are joined by a sequence of such switches: if M and N differ at a, write $\{a, b\} \in M$ and $\{a, c\} \in N$; switching the two M-edges through b and c creates $\{a, c\}$ and reduces the symmetric difference. This also gives an elementary proof of the divisibility clause in Proposition 2.1.

The switch calculation and the global pullback proof play different roles. The pullback proves the quotient for arbitrary matchings at once. The switch identity records how local changes of factorization move in the quotient space. A selected Coxeter orbit need not contain every perfect matching or every intermediate switch.

4 The rank-three configurations

We now select three finite matching orbits. This selection is discrete input: Proposition 2.1 applies to every matching, but it does not distinguish the Coxeter orbits.

For $q \in \{5, 7, 11\}$, index

$$\mathbb{P}^1(\mathbb{F}_q) = \{0, 1, \ldots, q-1, \infty\},$$

where a denotes $(1 : a)$ and ∞ denotes $(0 : 1)$. Let $G_q = \mathrm{PGL}_2(q)$ act in the usual way. The three base matchings are

T	q	M_T
A_3	5	$\{0, \infty\}, \{1, 4\}, \{2, 3\}$,
B_3	7	$\{0, 2\}, \{1, 4\}, \{3, \infty\}, \{5, 6\}$,
H_3	11	$\{0, 1\}, \{2, 5\}, \{3, 7\}, \{4, 9\}, \{6, 8\}, \{10, \infty\}$.

(4.1)

Define the *rank-three matching configuration*

$$\Omega_T = G_q \cdot M_T.$$

The stabilizers of the three displayed matchings are respectively S_4, S_4, A_5, and hence

$$|\Omega_{A_3}| = 5, \qquad |\Omega_{B_3}| = 14, \qquad |\Omega_{H_3}| = 22. \tag{4.2}$$

These marker spaces and their stabilizers are classical. In particular, Edge and Dye identify the exceptional finite conic configurations and substantial parts of their incidence geometry [1, 2]. We use the Coxeter names for the corresponding tetrahedral, octahedral, and icosahedral data; neither the orbit counts in (4.2) nor the ambiguity of an unmarked conic is asserted here as new.

Fix M_T as reference and apply (2.1) to every member of Ω_T. Write $R_d = \mathbb{F}_q[X, Y, Z]_d$ and

$$\Delta_Q = 4\partial_X\partial_Z - \partial_Y^2, \qquad \mathcal{H}_d = \ker(\Delta_Q : R_d \longrightarrow R_{d-2}). \tag{4.3}$$

The operator Δ_Q is the Laplacian attached to the dual quadratic form. In the degrees used below, $\mathcal{H}_d$ is the top harmonic summand, while powers of Q give radial summands.

Theorem 4.1 (The $3, 6, 10$ quotient ranks)**.** *For the matching configurations in (4.1), let*

$$W_T = \operatorname{span}_{\mathbb{F}_q}\{\Phi_{M_T}(M) : M \in \Omega_T\}.$$

Then

T	$\deg\Phi$	W_T	$\dim W_T$
A_3	1	$R_1 = \mathcal{H}_1$	3,
B_3	2	$R_2 = \mathcal{H}_2 \oplus \mathbb{F}_7 Q$	6,
H_3	4	$\mathcal{H}_4 \oplus \mathbb{F}_{11} Q^2$	10.

(4.4)

If $h_T = 4, 6, 10$ is the Coxeter number and $d = h_T/2 - 1$, the three rows have the uniform form

$$W_T = \begin{cases} \mathcal{H}_d, & d \text{ odd}, \\ \mathcal{H}_d \oplus \mathbb{F}_q Q^{d/2}, & d \text{ even}, \end{cases} \qquad \dim W_T = h_T - 1 + \mathbf{1}_{\{d \text{ even}\}}.$$

In particular, the first two configurations fill the complete quotient form space. The H_3 configuration spans a canonical ten-dimensional subspace of the fifteen-dimensional space R_4.

Proof mode. The quotient construction and the harmonic-space dimensions are proved symbolically. The assertions that the three displayed finite orbits span the stated subspaces are certificate-checked exact calculations, with a separate implementation that reconstructs the groups, matchings, quotient forms, Laplacian, and row reductions from the data in (4.1).

Proof. A matching in (4.1) has $m = 3, 4, 6$ edges, respectively, so Proposition 2.1 places its quotient in R_{m-2}, of degrees $1, 2, 4$. Representing the quotients in the standard monomial bases gives matrices with respectively $5, 14, 22$ rows. Exact row reduction over $\mathbb{F}_5, \mathbb{F}_7, \mathbb{F}_{11}$ gives ranks $3, 6, 10$.

It remains to identify the ten-space in the last row rather than record only its dimension. Direct differentiation gives

$$\dim \mathcal{H}_1 = 3, \qquad \dim \mathcal{H}_2 = 5, \qquad \dim \mathcal{H}_4 = 9.$$

The radial vectors $Q \in R_2$ and $Q^2 \in R_4$ are not harmonic in characteristics 7 and 11, respectively. The exact quotient matrices lie in

$$\mathcal{H}_1, \qquad \mathcal{H}_2 + \mathbb{F}_7 Q, \qquad \mathcal{H}_4 + \mathbb{F}_{11} Q^2.$$

Their computed ranks equal the dimensions $3, 6, 10$ of these spaces, so all three containments are equalities. Since $\dim R_4 = 15$, the last equality is proper. □

Corollary 4.2 (The exact H_3 loss)**.** *Over $\mathbb{F}_{11}$, the quartic form space has the Fischer decomposition*

$$R_4 = \mathcal{H}_4 \oplus Q\mathcal{H}_2 \oplus \mathbb{F}_{11} Q^2. \tag{4.5}$$

Consequently,

$$R_4 = W_{H_3} \oplus Q\mathcal{H}_2, \qquad R_4/W_{H_3} \simeq Q\mathcal{H}_2. \tag{4.6}$$

Equivalently,

$$W_{H_3} = \{f \in R_4 : \Delta_Q f \in \mathbb{F}_{11} Q\}. \tag{4.7}$$

Thus the H_3 quotient configuration retains the top harmonic nine-space and the deepest radial line, and omits precisely the five-dimensional middle radial layer.

Proof. For a homogeneous form f of degree r, direct differentiation gives

$$\Delta_Q(Qf) = (4r + 6)f + Q\Delta_Q f. \tag{4.8}$$

Hence $\Delta_Q(Qh) = 3h$ for $h \in \mathcal{H}_2$, while $\Delta_Q(Q^2) = 9Q$ over $\mathbb{F}_{11}$. Applying Δ_Q^2 and then Δ_Q shows that the three summands in (4.5) meet trivially. Their dimensions are $9, 5, 1$, so they exhaust the fifteen-dimensional space R_4. Theorem 4.1 then gives (4.6). If $f = h_4 + Qh_2 + cQ^2$ is its decomposition, then

$$\Delta_Q f = 3h_2 + 9cQ.$$

The quadratic decomposition $R_2 = \mathcal{H}_2 \oplus \mathbb{F}_{11} Q$ therefore gives (4.7). □

The rank calculation has two distinct boundaries. The common restriction and quotient are general consequences of the conic ideal, and (4.3) describes the intrinsic target subspaces. The fact that the three particular orbits in (4.1) attain those subspaces is finite. The primary certificate enumerates each full orbit and performs exact prime-field row reduction; the independent replay starts only from (4.1) and reconstructs the calculation without importing the primary implementation. Thus Theorem 4.1 does not infer a general rank formula from the three examples.

Equation (4.7) isolates a smaller conceptual target than the full rank calculation: the H_3 containment is the vanishing of the harmonic quadratic part of $\Delta_Q \Phi(M)$. Once that covariant identity is known, only spanning the resulting ten-space remains finite.

Corollary 4.2 identifies the linear information lost by the H_3 orbit, but linear span does not recover the two factorization sheets inside the finite configuration. Their recovery begins one categorical level higher, with products of affine evaluation functions.

5 Balanced sheets and cubic orientation

Show how second moments recover the two balanced halves in the B_3 and H_3 cases up to interchange. Identify degree three as the first signed moment that orients the halves.

6 Six profiles and matching-row reconstruction

Derive the two $1, 4, 6$ triples from subgroup marks, construct the six incidence profiles, and explain why their rank-two image still separates all six labels. End with the exact matching-row recovery statement.

7 Modular depth and arithmetic gluing

Present the modular depth quotient, the primitive $1 : 4 : 6$ relation, and the arithmetic splitting/gluing theorem as one change-of-characteristic story. Put the relative-cubic Tate plane in an appendix unless it shortens this argument.

8 Verification architecture

Distinguish conceptual proofs, classical inputs, Lean statements, and finite certificates. State the coordinate-to-Coxeter identifications and external semantic bridges explicitly.

A The relative-cubic Tate plane

Record the two canonical constructions and the proved boundary between the relative-cubic plane and the modular depth plane.

Conclusion

State exactly which factorization data the conic ideal remembers and which pointed or global data it still forgets.

References

[1] W. L. Edge, Conics and orthogonal projectivities in a finite plane, *Canadian Journal of Mathematics* **8** (1956), 362–382.

[2] R. H. Dye, Hexagons, conics, A_5 and $\mathrm{PSL}_2(K)$, *Journal of the London Mathematical Society* (2) **44** (1991), 270–286. doi:10.1112/jlms/s2-44.2.270.